

AI-Based Integrated Video Analytics System

Background

With the rapid growth of urban infrastructure, smart cities, transportation networks, and public surveillance systems, there is an increasing need for intelligent monitoring solutions that go beyond traditional CCTV surveillance. Manual monitoring of video feeds is inefficient, error-prone, and incapable of real-time decision-making.

Artificial Intelligence-based Video Analytics enables automated interpretation of video data to improve public safety, traffic management, crowd monitoring, and law enforcement operations.

Problem Description

Design and develop an **AI-Based Integrated Video Analytics System** capable of performing multiple intelligent video analytics functions through a **single unified platform**.

The proposed system should process video streams using either:

- **Edge Devices**, or
- **Server-based Software Solutions**,

while delivering real-time analytics and insights.

The solution must integrate multiple analytics modules into **one complete system set** instead of independent tools.

Required Analytics Modules

1. AI-Based Vehicle Counting & Classification

The system should:

- Detect vehicles from live or recorded video streams

- Count vehicles automatically
- Classify vehicle types (car, bike, bus, truck, etc.)
- Generate traffic analytics reports

2. Automatic Number Plate Recognition (ANPR)

The system should:

- Detect vehicle number plates
- Extract registration numbers using AI/OCR
- Store searchable vehicle records
- Support monitoring and enforcement applications

3. AI-Based Facial Recognition System (FRS)

The system should:

- Detect and recognize human faces
- Match faces against authorized/watchlist databases
- Enable identity-based monitoring while ensuring privacy considerations

4. People Counting & Gender Classification

The system should:

- Detect human presence
- Count number of individuals in real time
- Perform gender classification analytics
- Provide crowd density insights

Expected Solution

Participants are expected to develop:

- A **single integrated dashboard/system**
- Real-time video processing capability
- Edge or server deployment architecture
- Scalable and efficient AI models
- Data visualization and analytics reporting interface